



WORKSHEET 8

Student Name: Mohit Sharma

UID: 23BAI70733

Branch: CSE(3rd Year)

Section: 23AIT_Krg_G2

Semester: 5th

Date of Performance: 27/10/25

Subject Name: ADBMS

Subject Code: 23CSP-333

1. AIM:

Design a robust PostgreSQL transaction system for the students table where multiple student records are inserted in a single transaction.

If any insert fails due to invalid data, only that insert should be rolled back while preserving the previous successful inserts using savepoints.

The system should provide clear messages for both successful and failed insertions, ensuring data integrity and controlled error handling.

2. Tools Used : PostGres

Solutions:

Q1)

DROP TABLE IF EXISTS students;

```
CREATE TABLE students ( id
    SERIAL PRIMARY KEY,
    name VARCHAR(50), age
    INT, class INT
);
```

DO

\$\$ BE

GIN

BEGIN

INSERT INTO students(name, age, class) VALUES ('Supriya',16,8);

INSERT INTO students(name, age, class) VALUES ('Rakshit',17,8);

INSERT INTO students(name, age, class) VALUES ('Varun',19,9);

RAISE NOTICE 'Transaction Successfully Done';

EXCEPTION

WHEN OTHERS THEN

RAISE NOTICE 'Transaction Failed..! Rolling back changes.'; RAISE; END;

END;

\$\$;

SELECT * FROM students;

----- WRONG DATA TYPE SCENARIO -----

BEGIN; -- start transaction

SAVEPOINT sp1;

INSERT INTO students(name, age, class) VALUES ('Aarav',16,8);

SAVEPOINT sp2;

BEGIN

INSERT INTO students(name, age, class) VALUES ('Rahul','wrong',9); -- fails

EXCEPTION WHEN OTHERS THEN

RAISE NOTICE 'Failed to insert Rahul, rolling back to savepoint sp2';

ROLLBACK TO SAVEPOINT sp2; END;

-- Next insert

INSERT INTO students(name, age, class) VALUES ('Sita',17,10);

COMMIT; -- commit all successful inserts

3. Output:

Kargil Practicals/postgres@PostgreSQL 17

Query Query History Scratch Pad x

```

23      INSERT INTO students(name, age, class) VALUES ('Varun',19,9);
24
25      RAISE NOTICE 'Transaction Successfully Done';
26
27      EXCEPTION
28      WHEN OTHERS THEN
29          RAISE NOTICE 'Transaction Failed..! Rolling back changes.';
30          RAISE;
31      END;
32  END;
33  $$;
34
35  SELECT * FROM students;

```

Data Output Messages Notifications

NOTICE: Transaction Successfully Done
DO

Query returned successfully in 39 msec.

Kargil Practicals/postgres@PostgreSQL 17

Query Query History Scratch Pad x

```

28      WHEN OTHERS THEN
29          RAISE NOTICE 'Transaction Failed..! Rolling back changes.';
30          RAISE;
31      END;
32  END;
33  $$;
34
35  SELECT * FROM students;
36
37  -----WRONG DATA TYPE SCENARIO-----
38
39  BEGIN;
40
41  SAVEPOINT sp1;

```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	id [PK] integer	name character varying (50)	age integer	class integer
1	1	Supriya	16	8
2	2	Rakshit	17	8
3	3	Varun	19	9

```
Query Query History Scratch Pad x
42
43 SAVEPOINT sp2;
44 BEGIN
45     INSERT INTO students(name, age, class) VALUES ('Rahul','wrong',9); -- fails
46 EXCEPTION WHEN OTHERS THEN
47     RAISE NOTICE 'Failed to insert Rahul, rolling back to savepoint sp2';
48     ROLLBACK TO SAVEPOINT sp2;
49 END;
50
51 -- Next insert
52 INSERT INTO students(name, age, class) VALUES ('Sita',17,10);
53
54 COMMIT; -- commit all successful inserts
55
```

```
Data Output Messages Notifications
ERROR: syntax error at or near "INSERT"
LINE 2: INSERT INTO students(name, age, class) VALUES ('Rahul','...
      ^

SQL state: 42601
Character: 11
```

```
Data Output Messages Notifications
ROLLBACK

Query returned successfully in 35 msec.
```

4. Learning Outcomes:

- ❑ Understand the concept of PostgreSQL transactions and how to start, commit, and rollback.
- ❑ Learn how to use SAVEPOINT to handle partial rollbacks within a transaction.
- ❑ Practice controlled error handling for individual insert failures without affecting other successful operations.
- ❑ Gain experience in maintaining data integrity while performing multiple inserts.
- ❑ Learn to generate informative NOTICES to monitor transaction progress and errors.